

Program : Diploma in Civil and Environmental Engineering	
Course Code : 4351	Course Title: Soil Mechanics and Transportation Engineering
Semester : 4	Credits: 4
Course Category: Program core	
Periods per week: 4(L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge about index and engineering properties of soil and their relationships.
- To enable the students to explain the concepts of compressibility and bearing capacity of soil.
- To impart knowledge on the basics of highway geometric design and the pavement construction.
- To enable the students to distinguish the various transportation modes like railways and air ways.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of mathematics		Engineering Mechanics	2
Knowledge of construction techniques and construction materials.		Building Construction & Construction Materials	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Determine the index and engineering properties of soil	16	Applying
CO2	Determine the compressibility and bearing capacity of soil and illustrate the process of field	16	Applying

	investigation.		
CO3	Describe the highway geometric elements and outline the traffic studies and various traffic management aids.	16	Applying
CO4	Outline the different modes of transportation other than roadways.	12	Understanding

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3				3		
CO3	3				3		
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Determine the index and engineering properties of soil		
M1.01	Summarize the physical and index properties of soil and their relationships.	4	Applying
M1.02	Discuss the consistency limits of soil	2	Understanding
M1.03	Discuss the gradation and classification of soils.	3	Understanding
M1.04	Discuss the permeability of soil and explain the concept of Darcy's Law.	2	Understanding
M1.05	Outline the concepts of shear stress and Mohr-Coulomb failure theory and identify the suitable tests for evaluating shear strengths of various types of soil	5	Applying
Contents: Introduction to Soil Mechanics: Use of soil as construction material in Civil engineering. Physical and Index Properties of Soil: Soil as a three-phase system: define the terms -			

water content, void ratio, porosity, degree of saturation, density index, Unit weight of soil (bulk unit weight, dry unit weight, unit weight of solids, saturated unit weight, submerged unit weight).

Functional relationships: Relationship between void ratio and porosity; water content and void ratio; bulk unit weight and void ratio; dry unit weight and water content. Simple problems based on the above functional relationships.

Consistency limits of soil: Define the terms liquid limit, plastic limit and shrinkage limit, Plasticity index, consistency index, flow index and liquidity index. Simple problems.

Gradation of soil: Define coarse grained and fine-grained soils, effective diameter of soil, well graded soils, uniformly graded soils. BIS classification of soil based on particle size, procedure for determination of particle size distribution by sieve analysis.

Permeability of Soil: Define permeability. State Darcy's law of permeability, factors affecting permeability.

Shear Strength of Soil: Cohesion and internal friction, shear strength of soil, Mohr-Coulomb failure theory - Strength envelope, strength equation for purely cohesive and cohesion less soils. Procedure for determination of shear strength by direct shear test, unconfined compression test and vane shear test (simple problems on vane shear test only).

CO2	Determine the compressibility and bearing capacity of soil and illustrate the process of field investigation.		
M2.01	Discuss the different methods of compaction and explain the procedure for conducting Standard Proctor test for determining MDD and OMC.	2	Understanding
M2.02	Brief the different field compaction methods.	1	Understanding
M2.02	Outline the concept of consolidation and discuss the Terzaghi's theory of consolidation	3	Understanding
M2.03	Compare the processes of compaction and consolidation	1	Understanding
M2.04	Discuss about different site investigation methods.	3	Understanding
M2.05	Brief the different types of foundation and their suitability in different types of soil.	2	Understanding
M2.06	Describe the bearing capacity of soil and Terzaghi's bearing capacity theory.	4	Applying

Contents:

Compaction of Soil: Define compaction of soil. Procedure for Standard and Modified Proctor Test. Compaction curve, Optimum moisture content (OMC), maximum dry density (MDD) and Zero air voids line. Field methods of compaction - rolling, ramming and vibration.

Consolidation of soil: Define consolidation of soil and differentiate between compaction and consolidation. Describe Terzaghi's soil-spring analogy model to explain the process of consolidation and different stages of consolidation (Initial, primary and secondary

consolidation)

Site investigation and soil exploration: Define undisturbed samples. List the objectives of site investigation. Discuss the different stages of site investigation. Discuss the different methods of soil exploration - Open excavation methods, different types of boring and geophysical methods.

Different types of foundation - Shallow and Deep - isolated, strip, raft, Pile, well (Types and suitability).

Bearing Capacity of Soil: Define terms- bearing capacity, ultimate bearing capacity, safe bearing capacity and allowable bearing pressure. Terzaghi's theory of bearing capacity - introduction, assumptions, equation (derivation not required) and simple problems to find area and depth of foundation - Isolated and trip footing only (effect of water table not to be considered). Procedure for determination of bearing capacity of soil - Plate load test (as per IS:1888) and Standard Penetration Test (IS:2131).

CO3	Describe the highway geometric elements and outline the traffic studies and various traffic management aids.		
M3.01	Describe the IRC classification of roads and indicate the factors to be considered in alignment of a road.	2	Understanding
M3.02	Illustrate the components of a National Highway in embankment and summarize highway geometric design	5	Applying
M3.03	Outline the various traffic studies.	3	Understanding
M3.04	Illustrate various traffic management aids with sketches	2	Understanding
M3.05	Indicate the components of a pavement and different materials used for pavement construction	3	Understanding
M3.06	Compare the flexible and rigid pavements	1	Understanding

Contents:

Introduction to highway engineering and traffic engineering. Role of transportation in the development of nation, IRC classification of road, Major SH and NH in Kerala, Road alignment - Factors to be considered in road alignment.

Highway Geometrics: Standard cross-sections of national highway in embankment and cutting- various components, Camber, Gradient, Design speed, Sight distance (SSD): introduction to sight distance, reaction time, stopping sight distance, overtaking sight distance, intersection sight distance. Curves: Necessity, types: Horizontal, vertical curves - Extra widening of road, Super elevation:

Traffic Engineering: - Traffic volume studies, Traffic speed studies, OD studies, Intersections – types of road intersections, Types of interchanges, Road markings, Traffic signs - Types and purpose of signs.

Pavement materials- Stone aggregates: desirable properties, tests, requirements of aggregates for different types of pavements. Bituminous materials: types, tests on bitumen, desirable properties, selection of grade of bitumen. Pavement construction - Types of pavements, comparison of different types of pavements, functions of pavement components, construction of flexible pavement: water bound macadam, different types of bituminous pavements.			
CO4	Outline the different modes of transportation other than roadways.		
M4.01	Classify railway based on gauge	1	Understanding
M4.02	Illustrate the permanent way and its components	2	Understanding
M4.03	Summarize railway geometric features and classify station yards	2	Understanding
M4.04	Summarize points and crossings, signals and interlocking	2	Understanding
M4.05	Identify component parts of an airport and describe the various runway alignment patterns	3	Understanding
M4.06	Outline the terms- docks and harbours and brief their classification.	2	Understanding
Contents: Railway Engineering - Permanent way- requirements, components - Rail Gauge, types - Creep of rail: causes and prevention ,Sleepers, Ballast - functions and types - Rail fixtures and fastenings Railway Track Geometrics: Gradient, curves, grade compensation, super elevation, cant deficiency, negative cant, coning of wheel, tilting of rail, Track cross sections - standard cross section of single and double line broad gauge track in cutting and embankment, Points and crossings - Turn out - types, components. Station yards- types. Airport Engineering:- airport components - Runway, Apron, taxi way, terminal building, hangers, cargo, Fueling facility - layout of an airport, selection of site for airport Harbours – classification, features, requirements. Break waters- necessity and functions, classification. Docks – Functions and types - dry docks, wet docks (Introduction only)			

Text / Reference:

T/R	Book Title/Author
T1	Khanna S.K., Justo, C E G and Veeraragavan, A., “ <i>Highway Engineering</i> ”, Nem Chand and Brothers, Roorkee

T2	Punmia, B.C., “ <i>Soil Mechanics and Foundation Engineering</i> ”, Laxmi Publication, Delhi.
R3	Subramanian, K.P., “ <i>Highway, Railway, Airport and Harbour Engineering</i> ”, Scitech Publications, Hyderabad
R4	Arora, N. L., “ <i>Transportation Engineering</i> ”, Khanna Publishers, Delhi
R5	Subramanian, K.P., “ <i>Highway, Railway, Airport and Harbour Engineering</i> ”, Scitech Publications, Hyderabad.
R6	Murthy, V.N.S., “ <i>A Text book of Soil Mechanics and Foundation Engineering</i> ”, CBS Publishers & Distributors Pvt. Ltd., New Delhi.
R7	Saxena S C and Arora S P, “ <i>A Textbook of Railway Engineering</i> ”, Dhanpat Rai Publication
R8	Raj, P. Purushothama, “ <i>Soil Mechanics and Foundation Engineering</i> ”, Pearson India, New Delhi.
R9	Gopal Ranjan & A.S.R Rao, “ <i>Basic and Applied Soil Mechanics</i> ”, New Age International

Online resources:

Sl.No.	Website Link
1	https://nptel.ac.in/courses/105101087
2	https://archive.nptel.ac.in/courses/105/101/105101201/
3	https://nptel.ac.in/courses/105107123
4	https://nptel.ac.in/courses/105108069
5	https://archive.nptel.ac.in/courses/105/105/105105207/